Lab: Running a MySQL Database in a Docker Container

To Begin with the Lab

Summary

In this lab, we will learn how to run a MySQL database inside a Docker container using the official MySQL image available on Docker Hub.

Instead of installing the complete MySQL database engine directly on Ubuntu, we can use Docker to download the MySQL image and create a running container. We will map port 3306 so that applications and database clients can connect to the MySQL server.

We will also install the MySQL client in Ubuntu WSL and use it to connect to the MySQL database running inside the container.

2. Understanding MySQL in Docker

MySQL is a popular relational database management system (RDBMS) used to store and manage application data.

Normally, we install MySQL Server directly on a machine. However, Docker allows us to run MySQL inside a container without installing the complete database engine on our Ubuntu machine.

Docker provides an official MySQL image that we can use to create a MySQL container.

Important MySQL details:

Component	Description
Image	mysql
Default port	3306
Default username	root
Environment variable	MYSQL_ROOT_PASSWORD
Host address	127.0.0.1
Database engine	MySQL

Example:

Suppose we are developing a PHP application that needs a database.

Instead of installing MySQL Server directly on Ubuntu, we can run MySQL in a Docker container.

Our PHP application can then connect to the MySQL database through port 3306.

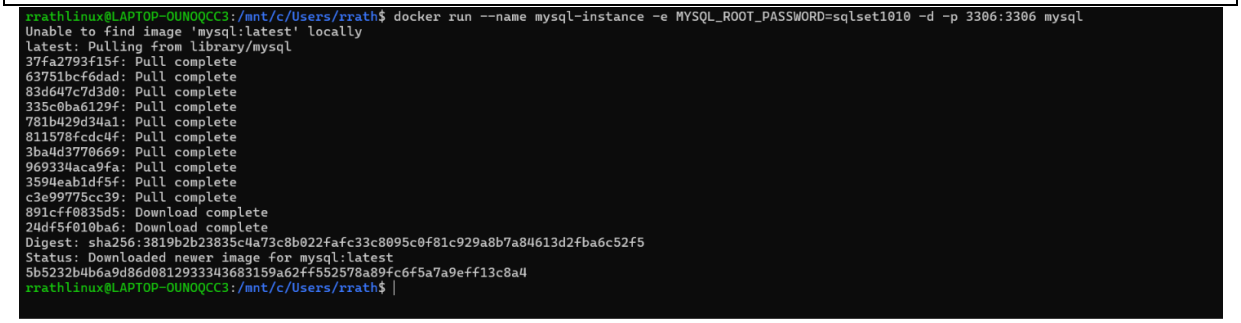
Lab Steps

Step 1: Run a MySQL Container

• Open your Ubuntu WSL terminal.

• Execute the following Docker command:

```
docker run --name mysql-instance -e MYSQL_ROOT_PASSWORD=sqlset1010 -d -p 3306:3306 mysql
```



Explanation:

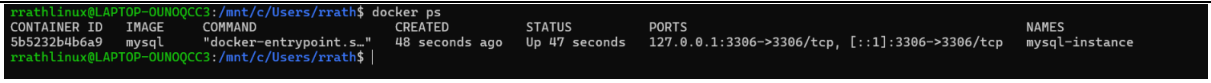
Command	Description
docker run	Creates and starts a container.
-d	Runs the container in detached mode.
--name mysql-container	Assigns a name to the container.
-p 3306:3306	Maps host port 3306 to container port 3306.
-e	Sets an environment variable.
MYSQL_ROOT_PASSWORD	Sets the MySQL root password.
mysql	Specifies the official MySQL image.

Note: The example password is for learning only. Use a strong, unique password for your own environment.

Step 2: Verify the MySQL Container

• Check whether the container is running:

```
docker ps
```



• To see all containers, including stopped containers:

```
docker ps -a
```

```
rrathlinux@LAPTOP-OUNOQCC3:/mnt/c/Users/rrath$ docker ps -a
CONTAINER ID   IMAGE     COMMAND                  CREATED              STATUS              PORTS                                                  NAMES
5b5232b4b6a9   mysql     "docker-entrypoint.s…"   About a minute ago   Up About a minute   127.0.0.1:3306->3306/tcp, [::1]:3306->3306/tcp   mysql-instance
rrathlinux@LAPTOP-OUNOQCC3:/mnt/c/Users/rrath$
```

• To view the container logs:

```
docker logs mysql-container
```

Wait until MySQL finishes its initial setup before connecting.

Step 3: Update Ubuntu Package Index

To install the MySQL client in Ubuntu, first update the package index.

sudo apt update

This command refreshes the package information available to Ubuntu.

Step 4: Install MySQL Client

• Install the MySQL client tools:

```
sudo apt install mysql-client -y
```

```
rrathlinux@LAPTOP-OUNOQCC3:/mnt/c/Users/rrath$ sudo apt install mysql-client -y
[sudo: authenticate] Password:
Installing:
  mysql-client

Installing dependencies:
  libgoogle-perftools4t64  libtcmalloc-minimal4t64  libunwind8  mysql-client-core  mysql-common

Summary:
  Upgrading: 0, Installing: 6, Removing: 0, Not Upgrading: 72
  Download size: 2612 kB
  Space needed: 49.6 MB / 1024 GB available

Get:1 http://archive.ubuntu.com/ubuntu resolute/main amd64 libtcmalloc-minimal4t64 amd64 2.18.1-1 [138 kB]
Get:2 http://archive.ubuntu.com/ubuntu resolute/main amd64 libunwind8 amd64 1.8.3-0ubuntu1 [59.6 kB]
Get:3 http://archive.ubuntu.com/ubuntu resolute/main amd64 libgoogle-perftools4t64 amd64 2.18.1-1 [199 kB]
Get:4 http://archive.ubuntu.com/ubuntu resolute-updates/main amd64 mysql-client-core amd64 8.4.11-0ubuntu0.26.04.1 [2200 kB]
Get:5 http://archive.ubuntu.com/ubuntu resolute/main amd64 mysql-common all 5.8+1.1.1ubuntu2 [7002 B]
Get:6 http://archive.ubuntu.com/ubuntu resolute-updates/main amd64 mysql-client amd64 8.4.11-0ubuntu0.26.04.1 [9126 B]
Fetched 2612 kB in 11s (231 kB/s)
Selecting previously unselected package libtcmalloc-minimal4t64:amd64.
(Reading database ... 37477 files and directories currently installed.)
Preparing to unpack .../0-libtcmalloc-minimal4t64_2.18.1-1_amd64.deb ...
Unpacking libtcmalloc-minimal4t64:amd64 (2.18.1-1) ...
Selecting previously unselected package libunwind8:amd64.
Preparing to unpack .../1-libunwind8_1.8.3-0ubuntu1_amd64.deb ...
Unpacking libunwind8:amd64 (1.8.3-0ubuntu1) ...
Selecting previously unselected package libgoogle-perftools4t64:amd64.
Preparing to unpack .../2-libgoogle-perftools4t64_2.18.1-1_amd64.deb ...
Unpacking libgoogle-perftools4t64:amd64 (2.18.1-1) ...
Selecting previously unselected package mysql-client-core.
Preparing to unpack .../3-mysql-client-core_8.4.11-0ubuntu0.26.04.1_amd64.deb ...
Unpacking mysql-client-core (8.4.11-0ubuntu0.26.04.1) ...
Selecting previously unselected package mysql-common.
Preparing to unpack .../4-mysql-common_5.8+1.1.1ubuntu2_all.deb ...
Unpacking mysql-common (5.8+1.1.1ubuntu2) ...
Selecting previously unselected package mysql-client.
Preparing to unpack .../5-mysql-client_8.4.11-0ubuntu0.26.04.1_amd64.deb ...
Unpacking mysql-client (8.4.11-0ubuntu0.26.04.1) ...
Setting up mysql-common (5.8+1.1.1ubuntu2) ...
update-alternatives: using /etc/mysql/my.cnf.fallback to provide /etc/mysql/my.cnf (my.cnf) in auto mode
Setting up libunwind8:amd64 (1.8.3-0ubuntu1) ...
```

Understanding:

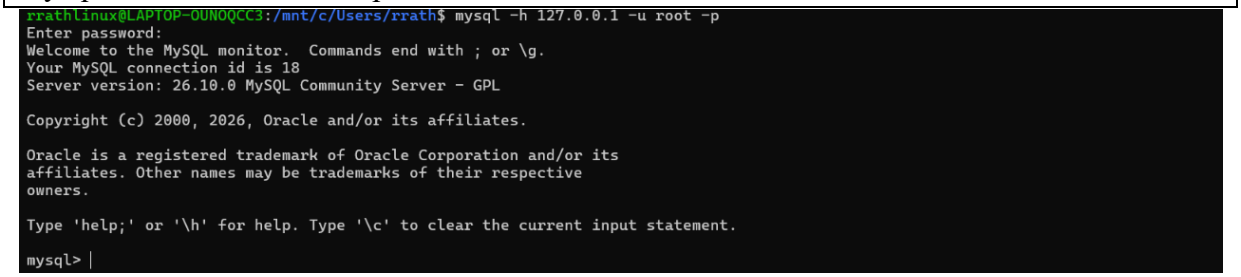
The MySQL client allows us to connect to a MySQL database server.

We do not need to install the complete MySQL Server in Ubuntu because our database engine is already running inside the Docker container.

Step 5: Connect to the MySQL Container

• Execute the following command:

```
mysql -h 127.0.0.1 -u root -p
```



Explanation:

Option	Description
mysql	Starts the MySQL client.
-h	Specifies the database host.
127.0.0.1	Loopback address of the local machine.
-u root	Connects using the root username.
-p	Prompts for the password.

• Enter the password configured when creating the container:

SQLset1010

Note: The password will not appear on the screen while typing. This is normal.

If the connection is successful, you will see the MySQL prompt:

mysql>

You are now connected to the MySQL database engine running inside the Docker container.

Step 6: Exit the MySQL Client

• To exit the MySQL command-line interface:

exit;



You will return to your Ubuntu terminal